Hendra virus (HeV) was first described in 1994 following the outbreak of a novel disease fatally affecting horses and humans in south-east Queensland. Sporadic outbreaks in Queensland and north-east NSW were identified in 1999, 2004, 2006, 2007, 2008, 2009 & 2010. 2011 saw an unprecedented number of new cases occur. To date in 2012 there have been eight incidents. In total since 1994 there have been 77 incidents. There have been seven known human cases including four deaths.

Fruit bats (flying foxes) have been identified as the natural host of the virus.

The inability to efficiently identify variants circulating in the natural host poses a significant constraint to the development of sensitive diagnostic tests, to response preparedness, and to risk management strategies.

In June 2008, Queensland received funding from the Australian Government Department of Agriculture, Fisheries and Forestry (DAFF) as part of the Wildlife Emergency Disease Preparedness Program (WEDPP), for Stage 1 of an investigation of HeV strain diversity in bats. That study design incorporated both our better understanding of HeV infection dynamics in bats, and sampling and diagnostic approaches to maximise test sensitivity. Our efforts that year yielded the first ever identification of HeV genome in flying fox urine.

In 2009 Queensland received joint WEDPP and Australian Biosecurity Cooperative Research Centre (AB CRC) funding to continue the project, detecting HeV genome in 37 of over 1000 pooled urine samples collected, allowing preliminary comparative phylogenetic analyses of HeV sequence from bat, horse and human, as well as providing insight into the temporal and spatial pattern of HeV infection in flying foxes. Our long-standing collaboration with Dr Linfa Wang's group at the CSIRO Australian Animal Health laboratory (AAHL) resulted in the isolation of Hendra virus from PCR-positive samples on multiple occasions. The project has also yielded two new (yet to be characterised) paramyxoviruses in flying fox urine.

In 2010 this project was successful in obtaining WEDPP funding to expand investigation of Hendra virus strain diversity in flying foxes to include targeted screening of feral horse and feral pig populations. The WEDPP funding will be augmented by legacy AB CRC funds, as well as in-kind contributions by Biosecurity Queensland and AAHL.

In this proposal, the key research questions are 'What is the diversity of Hendra viruses occurring in Australia?' and 'What is the spatio-temporal pattern and frequency of infection in flying fox populations'?

Project objectives

- Identification of the presence or absence of HeV infection in high-risk feral pig and horse populations
- A more complete phylogenetic analysis of Hendra viruses circulating in flying foxes, with this year's surveillance targeting previously unsampled locations
- Increased certainty regarding the spatio-temporal pattern of HeV infection in flying foxes

Background

There have been 14 identified spillovers of Hendra virus since 1994, resulting in more than 40 recognised equine cases (75% case fatality rate) and 7 human cases (>50% case fatality rate). Fruit bats (Pteropus species, commonly known as flying foxes) have been identified as the natural host of the virus.

The inability to efficiently identify Hendra virus variants circulating in the natural host has been a significant constraint to the development of sensitive diagnostic tests, to response preparedness, and to risk management strategies.

In 2008 we received WEDPP funding for Stage 1 of an investigation of HeV strain diversity in flying foxes. The study design incorporated both our better understanding of HeV infection dynamics in bats, and sampling and diagnostic approaches to maximise test sensitivity. Our efforts that year yielded the first ever identification of HeV genome in flying fox urine. In 2009 we received further WEDPP and Australian Biosecurity CRC funding to complete Stage 2 of the project, detecting HeV genome in 37 of over 1000 pooled urine samples collected, allowing comparative phylogenetic analyses of HeV sequence from bat, horse and human, as well as providing insight into the temporal and spatial pattern of HeV infection in bats. In addition to this our long-standing collaboration with Dr Linfa Wang's group at AAHL resulted in the isolation of Hendra virus from PCR-positive samples on multiple occasions, and the isolation of two new (yet to be characterised) paramyxoviruses from flying fox urine.

The current Stage 3 of the project seeks to expand our investigation of Hendra virus strain diversity in flying foxes to include targeted screening of feral horse and feral pig populations in addition to ongoing surveillance of targeted flying fox populations..

Thus, in addition to the continuing key research questions

- What is the diversity of Hendra viruses occurring in Australia?
- What is the spatio-temporal pattern and frequency of infection in flying fox populations?
  we add the following research questions
- Is there evidence of Hendra virus infection in feral pig populations in Queensland? and relatedly
- Is there evidence of Hendra virus infection in feral horse populations in Queensland?

The key project deliverables are

- Identification of the presence or absence of HeV infection in high-risk feral pig and horse populations.
- A more complete phylogenetic analysis of Hendra viruses circulating in flying foxes, with this year's surveillance targeting previously unsampled locations.
- Increased certainty regarding the spatio-temporal pattern of HeV infection in flying foxes.

These deliverables build on our understanding of the ecology of Hendra virus, and complement current and proposed laboratory-based projects, and disease modeling projects.

Source: https://www.agriculture.gov.au/biosecurity-trade/policy/emergency/wedpp/hendra-mapping